TOML-MIRI: Convex Optimization

Jose M. Barceló Ordinas, Jorge García Vidal, and Pau Ferrer Cid

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These Lecture Notes are mainly based on the book "Convex Optimization" (`https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf`) by S. Boyd and L. Vanderberghe.

Video lectures for this course can be found in: (`https://www.youtube.com/playlist?list=PLoROMvodv4rMJqxxviPa4AmDClvcbHi6h`).

1 Introduction: Unconstrained minimization of smooth functions

We are interested in finding minimum values of multivariate scalar functions. Let us start with two simple examples, in which the minimization is done through the whole domain of the function (i.e. this is an unconstrained minimization problem) of differentiable functions.

Our first example is:

$$min_x \; c^T x,$$

which consists is finding the minimum value of a plane. Unless the plane is horizontal (i.e. $c^t = 0$), problem has no optimal value x^*, as it is **unbounded below**. If $c = 0$, any point is a solution, and the optimal value is $p^* = 0$.

Our second example is:

$$min_x \; \frac{1}{2} x^T P x + q^T x,$$

where P is a square matrix. Without loss of generality, we can assume that P is symmetric, as $x^T P x = \frac{1}{2} x^t (P + P^T) x$, and $\frac{1}{2}(P + P^T)$ is indeed symmetric.

Assume that u, λ are an eigenvector/eigenvalue pair of matrix P. If x is aligned with u, for instance $x = tu$ for some scalar t, the function takes values $\lambda t^2 + bt$, where $b = q^t u$. If $\lambda > 0$ this function has a minimum value at $t = -\frac{b}{2\lambda}$. If $\lambda = 0$ this function is unbounded below unless $b = 0$. If $\lambda < 0$ this function is unbounded below.

On the other hand, we can find a simple equation for the points for which the gradient vanishes, obtaining:

$$Px = -q.$$

As the function is differentiable, if x is a local minima, the gradient must vanish there (although the reverse is not true).

Our conclusion is that if all the eigenvalues of P are positive (in other words, P is **positive definite**) the problem has an single optimal solution, given by the solution of the previous linear system of equations, i.e. $x^* = -P^{-1} q$.

If some eigenvalues are zero, but no eigenvalue is negative, in other words P must be **semidefinite positive**, we must impose $q = 0$ (otherwise the function is unbounded below), and in this case the minimum value is 0, and it is achieved at the subspaces generated by linear combinations of the eigenvalues associated with zero eigenvalues.

Otherwise, the function is unbounded below.

In these two examples, we were able to find the optimal solutions, but in the general case:

$$min_x\ f_0(x),$$

finding the solution becomes very difficult, as the function can have several local minima, saddle points, etc. This problem can be even more difficult if we impose **constraints** for the solution:

$$min_x\ f_0(x)$$

$$subject\ to\ x \in C.$$

1.1 The role of convexity

There is an special case where there are efficient algorithms for finding the solution of the previous problem in most practical cases, which correspond to the case in which $f(x)$ is a **convex function**, and C is a **convex set**. Moreover, there are available software packages with efficient implementation of these algorithms.

In summary, if you formulate your problem in terms of a convex optimization problem, you can expect to find a solution with good accuracy with a reasonable amount of computation. On the other hand, many practical problems can be formulated in terms of a convex problem, meaning that is very interesting to be able to identify convex optimization problem.

2 Convex sets, convex functions and convex optimization problems

Let C be a subset of R^n. C is a **convex set** when for all x and y in C we have that $\theta x + (1-\theta)y \in C$ for $0 \leq \theta \leq 1$.

Let f be a real function $f : Dom(f) \subset \mathbf{R}^n \rightarrow \mathbf{R}$. We say that f is a **convex function** when **its domain is a convex set** and, for all $x, y \in Dom(f)$, we have: $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$ for $0 \leq \theta \leq 1$.

A function f is **concave** when $-f$ is convex.

An alternative definition of convexity is based on the notion of the **epigraph** of a function, which is the subset of R^{n+1} defined by $Epi(f) = \{(x, y) : x \in Dom(f) \subset \mathbf{R}^n, y \in \mathbf{R}, y \geq f(x)\}$.

We say that **a function f is convex when its epigraph is a convex subset of $\mathbf{R}^{n+1}$**.

In this course we are interested in **constrained convex optimization problems** of the form:

$$minimize\ f_0(x),\ subject\ to\ x \in C,$$

where f_0 is a convex objective function and C is a convex set. We define the **feasible set** F as: $F = C \cup Dom(f_0)$, which is also a convex set.

If $C = Dom(f_0)$, we will say that the optimization problem is **unconstrained**.

If $F = \emptyset$ we say that the problem is **unfeasible**.

If for a feasible problem, there is no $x^* \in F$ such that $\forall x \in F, f_0(x^*) \leq f_0(x)$ we say that the problem is **unbounded below**

2.1 Expressing a constrained minimization problem as an unconstrained problem

Assume that for a set $F \subset \mathbf{R}^n$, we define the **indicator function** $I_F(x)$ as:

$$I_F(x) = 0 \ \ if \ \ x \in F,$$

$$I_F(x) = \infty \ \ if \ \ x \notin F.$$

Then a constrained optimization problem with feasible set F can be written as an unconstrained optimization problem:

$$min_x \quad f_0(x) + I_F(x).$$

3 Set transformations that preserve convexity

Very often a set is defined as transformations of other sets that are known to be convex. It is thus very important to know which transformations preserve convexity.

3.1 The intersection of convex sets is a convex set

Assume that $L, M \subset \mathbf{R}^n$ are convex sets, this means that every convex combination of two elements $x, y \in L \bigcap M$ belongs to L and M, meaning that also belongs to $L \bigcap M$, thus proving that **the intersection of convex sets is convex**.

3.2 The affine mapping of a convex set is a convex set

Assume that $L \subset \mathbf{R}^n$ is a convex set. We define $N \subset \mathbf{R}$, as an affine mapping of L, i.e. every element $x \in L$ maps into an element $z \in N$ as $z = Ax + d$, for a constant matrix A and a constant vector d. We will use the notation $N = AL + d$.

Assume $z, z' \in N$ with $z = Ax + d$ and $z' = Ax' + d$, with $x, x' \in L$. Then convex combinations of z, z' are of the form $\theta z + (1-\theta)z' = A[\theta x + (1-\theta)x'] + d$, where $\theta x + (1-\theta)x' \in L$, meaning that these convex combinations belong to N, implying that N is a convex set.

For instance, **displacements, scaling, sums, and projections of convex sets are convex**.

3.3 The convex combination of convex sets is a convex set

Using a similar reasoning we see that if $N = \theta L + (1-\theta)M$ for L, M convex sets and $0 \leq \theta \leq 1$, N is also a convex set.

3.4 Sublevel sets of convex functions

If $f(x)$ is a convex function, then the set of x that fulfill:

$$f(x) \leq L,$$

is also a convex set (sublevel set).

Note: If f is convex we cannot say anything about the convexity of $f(x) \geq L$ or $f(x) = L$. **For guaranteeing convexity, the inequality must be "lower than"**.

4 Important examples of convex sets

A fundamental step in solving an optimization problem is in determining whether the feasible set is or not a convex set. It is thus very important to be able to identify the convexity of some sets that very often appear in applications.

4.1 Linear and affine subspaces are convex sets

A subset $L \subset \mathbf{R}^n$ is a *linear subspace* when for all $a_1, a_2 \in L$ we have that any *linear combination* also belongs to L: $x_1 a_1 + x_2 a_2 \in L$; $x_1, x_2 \in \mathbf{R}$.

A subset $L \subset \mathbf{R}^n$ is a *affine subspace* when for all $a_1, a_2 \in L$ we have that any *affine combination* also belongs to L: $x_1 a_1 + x_2 a_2 \in L;\ x_1, x_2 \in \mathbf{R},\ x_1 + x_2 = 1$.

Using this definition it is trivial to prove that **linear and affine subspaces are convex sets.**

It is obvious than R^m is a convex set, which means that $\{Ax \in R^n; x \in R^m\}$ and $\{Ax + b \in R^n; x \in R^m\}$ are also convex sets (of dimension n).

We also have that the **affine hull** $\{Ax + b \in R^n; x \in R^m, \mathbf{1}^T x = 1\}$ is a convex set (of dimension $m - 1$).

4.2 This means that $Ker(A) = \{x \in R^m;\ Ax = 0\}$ is a convex set. $\{x \in R^m;\ Ax = b\}$ is also convex

Obvious as $Ker(A)$ is a linear subspace.

4.3 The positive orthant $\{x \in R^m; x \geq 0\}$ is a convex set. This implies that cones are also convex sets

The set $\{x \in R^m; x \geq 0\}$ is convex, meaning that the **cone** $\{Ax; x \geq \mathbf{0}\}$ is a convex set.

Note: In this equation and the rest of LNs, we assume that *the inequality sign applied to vectors means component-wise inequalities.*

4.4 Simplexes $\{x \in R^m; \mathbf{1}^T x = 1,\ x \geq \mathbf{0}\}$ are convex sets. This implies that segments and convex hulls are also convex sets

A **simplex** $\{x \in R^m; \mathbf{1}^T x = 1,\ x \geq \mathbf{0}\}$ is a convex set.

Assume $A = (a_1, ..., a_m)$. Then the **convex hull** of $\{a_1, ..., a_m\}$, defined as $\{Ax; \mathbf{1}^T x = 1,\ x \geq \mathbf{0}\}$ is a convex set.

The case $m =$ corresponds to a **segment** between a_1 and a_2.

4.5 Hyperplanes, halfspaces

As we have seen, way of defining linear or affine subspaces is by means of the kernel of a linear mapping. The simplest form is the definition of a **hyperplane**:

$$\{x;\ a^T x = 0\},$$

which is a linear subspace of dimension $n-1$. Analogously we can define an **affine hyperplane**:

$$\{x;\ a^T(x-x_0)=0\},$$

or equivalently ($a^T x_0 = b$):

$$\{x;\ a^T x = b\}.$$

We can define a **halfspace** as:

$$\{x;\ a^T x \geq b\}.$$

4.6 Polyhedras and polytopes

The intersection of halfspaces is a **polyhedra**:

$$\{x;\ Ax \geq b\}.$$

The intersection of a polyhedra with an affine subspace is also a polyhedra:

$$\{x;\ Ax \geq b, Cx = d\}.$$

A **polytope** is a bounded polyhedra (there are different definitions for this)

4.7 Paraboloids and ellipsoids

The set

$$\{x;\ \frac{1}{2}x^T Qx + p^T x + c \leq 1\},$$

where Q is a positive definite matrix, i.e. $Q \in S_n^{++}$, p is a vector and c is an scalar, is an ellipsoid. Ellipsoids are convex sets. The same convexity appears when P is positive semi-definite.

If Q is non-symmetric, we can substitute it by its symmetric part $S = \frac{1}{2}(Q+Q^T)$ and the quadratic form is unchanged:

$$\frac{1}{2}x^T(Q+Q^T)x = \frac{1}{2}x^T Qx + \frac{1}{2}x^T Q^T x = \frac{1}{2}x^T Qx + \frac{1}{2}x^T Qx = x^T Qx.$$

Another way of writing an ellipsoid is by $\{x_c + Au \mid ||u||_2 = 1\}$, where A can be rectangular and either row/column full norm (depending on whether is fat-short or tall-thin)

4.8 The norm balls

For any norm, the ball $\{x : | \, ||x - x_0|| \leq r\}$ is a convex set.

Note that norms fulfill the triangular inequality: $||y + w|| \leq ||y|| + ||w||$:

$$||x_1 a_1 + x_2 a_2|| \leq x_1 ||a_1|| + x_2 ||a_2||,$$

as $x_1, x_2 \geq 0$, meaning that $||.||$ is a convex functions. Sublevel sets of convex functions are convex sets.

4.9 The cone of positive definite matrices

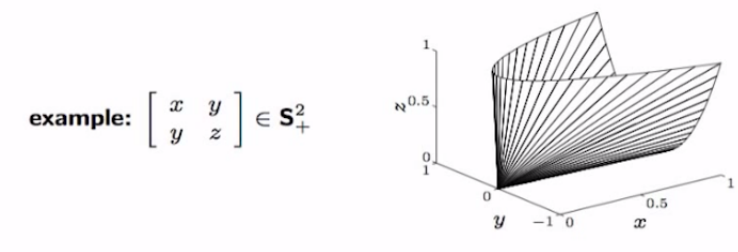
A symmetric matrix $Q \in \mathbf{R}^{n \times n}$ is *positive (semi)definite* when for all vectors x we have $x^T Q x \geq 0$. For positive definite, the equality is obtained only when $x = 0$.

S_n^+ is the *set of positive semi-definite matrices, PSD*, and it is a convex set. This is easily seen by considering a convex combination of $P, Q \in S_n^+$:

$$x^T(\theta P + (1 - \theta)Q)x = \theta x^T P x + (1 - \theta) x^T Q x \geq 0.$$

Using a similar argument with a conic combination we can easily see that this set is in fact a cone.

We use the notation S_n^{++} for the *set of positive definite matrices, PD*, which is also a convex set and a cone.



5 Separating and support hyperplanes

If C and D are non overlaping convex sets, there exists a *separating hyperplane* $a^T x = b$ that fulfills:

$$a^T x \geq b; \; \forall x \in C,$$

$$a^T x \leq b; \; \forall x \in D.$$

If C is a convex set, and x_o belongs to the boundary of C, there is a vector a^T for which:

$$a^T x_0 \geq a^T x;\ \forall x \in C.$$

$a^T(x - x_0) = 0$ defines the *supporting hyperplane*.

6 Basic properties of convex functions

Let f be a real function $f : Dom(f) \subset \mathbf{R}^n \rightarrow \mathbf{R}$. Recall that we say that f is a **convex function** when its domain is a convex set and for all x and y in $Dom(f)$ we have: $f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$ for $0 \leq \alpha \leq 1$.

6.1 Epigraphs of convex functions are convex sets

An alternative definition of convexity of functions is based on the notion of the **epigraph** of a function, which is the subset of R^{n+1} defined by $epi(f) = \{(x, y) : x \in Dom(f) \subset \mathbf{R}^n, y \in \mathbf{R}, y \geq f(x)\}$.

We say that **a function f is convex if and only if (iff) its epigraph is a convex subset of $\mathbf{R}^{n+1}$**.

6.2 For convex functions, local minima are also global minima

The most important property of convex functions is that the **local minima are also global minima**.

6.3 Convex functions are also convex restricted to any ray from a point of the domain

Let f be a real function $f : Dom(f) \subset \mathbf{R}^n \rightarrow \mathbf{R}$. f is convex *iif* the function $g : Dom(g) \subset \mathbf{R} \rightarrow \mathbf{R}$ defined as $g(t) = f(x + tv)$, with domain:

$$Dom(g) = \{t;\ x + tv \in Dom(f), x \in Dom(f), v \in \mathbf{R}^n\}$$

is convex.

7 Characterization of convex functions for the differentiable case

Let f be defined in an *open domain* $Dom(f)$.

We say that f is *differentiable iff the gradient* ∇f *exists and is continuous at all points of* $Dom(f)$.

We say that f is *twice differentiable iff the Hessian* H_f *exists and is continuous at all points of* $Dom(f)$.

7.1 Gradient approximations are always lower bounds of convex differentiable functions

When f is a differentiable function defined in a convex and open set $Dom(f)$, then a *necessary and sufficient* condition for f to be a convex function is:

$$f(y) \geq f(x) + \nabla f(x)^T (y - x),$$

for all y and x in $Dom(f)$.

7.2 Hessians of twice differentiable convex functions are PSD

When f is a twice differentiable function defined in a convex and open set $Dom(f)$, then a *necessary and sufficient* condition for f to be a convex function is that:

$$H_f(x) \in S_n^+,$$

for all x in $Dom(f)$.

8 Examples of convex functions

8.1 Functions on R

- $f(x) = e^{ax}$ is convex.
- $f(x) = -\log x$ is convex.
- $f(x) = x^a,\ x \in \mathbf{R}^+$ is concave for $0 \leq a \leq 1$, convex otherwise.
- $f(x) = x \log x$ is convex.

8.2 Functions on $\mathbf{R}^n$

- $f(x) = ||x||_p$ is convex (e.g. $p = 1, 2, \infty$) (proof: use triangular inequality).
- $f(x) = \max_i(x_1, ..., x_n)$ is convex (proof: this is the intersection of epigraphs of functions $f_i(x) = x_i$).

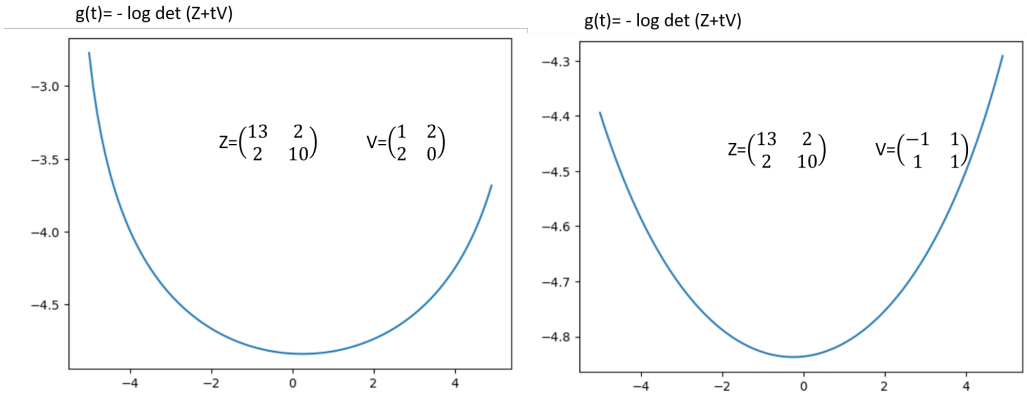
- $f(x) = -\log(\sum_i e_i^x)$ is convex, (proof: The hessian H_h is PSD, see VB's book, pp 74).

8.3 Functions on $\mathbf{R}^{n\times n}$

- $f(X) = -\log\det(X)$, where $X \in S_n^{++}$, is convex.

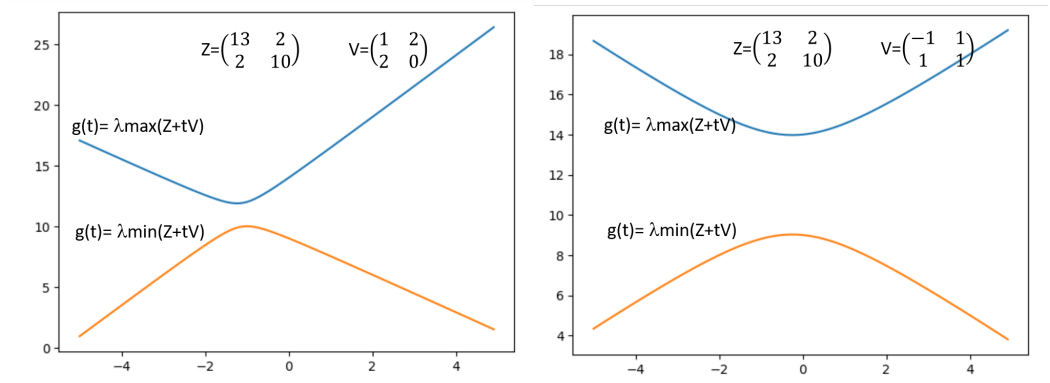
  Assume $Z \in S_n^{++}$. We consider the ray $Z+tV$, where V is a symmetric matrix (but in general not a PD), for values of t such that $Z+tV \in S_n^{++}$. We use $Z = Z^{1/2}Z^{1/2}$ to obtain $Z+tV = Z^{1/2}(I+tZ^{-1/2}VZ^{-1/2})Z^{1/2}$, that leads to: $-\log det(Z+tV) = -\log detZ - \log det(I+tZ^{-1/2}VZ^{-1/2}) = -\log detZ - \sum_i \log(1+t\lambda_i)$, where λ_i are the real eigenvalues of matrix $Z^{-1/2}VZ^{-1/2}$. The second derivative respect to t is $\sum_i \frac{\lambda_i^2}{(1+t\lambda_i)^2}$, which is positive in all cases.

  Note: If X is the variance-covariance matrix of a Gaussian random variable, $\log\det(X)$ is proportional to the volume of confidence ellipsoids. This functions appears frequently in optimization problems.



- The maximum eigenvalue of a symmetric matrix $X \in S_n$, $\lambda_{max}(X)$ is a convex function of A. The minimum eigenvalue $\lambda_{min}(X)$ is concave.

  The proof is surprisingly easy: the function a^TXa of X for fixed a is linear. We can define then a family of convex functions taking different values of a with $||a|| = 1$. $\lambda_{max}(X) = \sup_{||a||=1} a^TXa$, and the supremum of a family of convex functions is convex (see next section). Analogously, $\lambda_{min}(X) = \inf_{||a||=1} a^TXa$, and the infimum of a family of concave functions is concave (see next section), and linear functions are both convex and concave.



9 Operations that preserve convexity

9.1 Affine map

If f is convex, $x \in C$, a convex set, then $f(Ax + b)$ is convex. If f is concave, $f(Ax + b)$ is concave.

- Example: if $x \in R^n$, then $||Ax + b||_2^2$ is a convex function of x (least squares).

9.2 Non-negative weighted sums and integrals

If f_i are convex functions, and $\theta_i \geq 0$, then $\sum_i \theta_i f_i(x)$ is convex.

If $f(x, t)$ is a convex function on x for every fixed value of t, and $p(t) \geq 0 \ \forall t$, then $\int_t p(t) f(x, t)\, dt$ is convex.

9.3 Pointwise maximum/supremum

- If f_i are convex functions, then the point-wise maximum $g(x) = max_i(f_1(x), ...,)$ with $Dom(g) = \bigcap Dom(f_i)$ is convex.
- If $f(x, y)$ is convex in x (i.e. y act as a continuous index i), then $g(x) = sup_y(f(x, y))$ is convex.

Note that these two properties can be expressed as an intersection of epigraphs.

Analogoulsy, the infimum of concave functions is concave (as it is the supremum of $-f$). This is going to be an important property when we discuss the **Lagrange dual function**.

9.4 Optional: Pointwise minimum/infimum

If $f(x, y)$ is convex in x *and also in* y, and C is a convex set, then $inf_{\{y \in C\}}\ f(x, y)$ is concave in x.

NOTE: $f(x, y) = xy$ is convex on x (i.e. for fix y) and convex on y (for fixed x). However *this is not a convex function*. Example $f((0, 0) + t(1, -1)) = -t^2$ which is non convex.

9.5 Composition (scalar case)

Let $f = h(g(x))$. For simplicity, we assume $x \in \mathbf{R}$. We have:

$$f''(x) = h''(x)(g'(x))^2 + h'(x)g''(x),$$

meaning that if both terms are positive, f is convex (and if both terms are negative, f is concave). For instance:

- h is convex and non-decreasing and g is convex, implies f is convex.
- h is convex and non-increasing, and g is concave, implies f is convex,

etc.

Examples:

- If $g(x)$ is convex, then $exp(g(x))$ is convex
- $g(x)$ is concave and positive, then $\log(g(x))$ is convex.
- $g(x)$ is convex and positive, then $1/g(x)$ is convex,

etc

These rules also apply even when the functions h or g are not differentiable.

If $Dom(h)$ is not $\mathbf{R}$, we must use the extended version of h, i.e. $\hat{h}$, in the non-increasing or non-decreasing conditions, for instance

- h is convex, $\hat{h}$ is non-decreasing and g is convex, implies f is convex.
- h is convex, $\hat{h}$ is non-increasing, and g is concave, implies f is convex,

etc.

9.6 Composition (vector case)

Let $f = h(g_1(x), ..., g_p(x))$. For simplicity we assume $x \in \mathbf{R}$. We have:

$$f''(x) = g'(x)^T H_h(g(x))g'(x) + \nabla h(g(x))g''(x),$$

meaning that if both terms are positive, f is convex (and if both terms are negative, f is concave). For instance:

- h is convex and non decreasing in all coordinates and g_i are convex, implies f is convex.
- h is convex and decreasing in all coordinates, and g_i are concave, implies f is convex, etc

Again, these rules also apply even when the functions h or g are not differentiable.

For instance $f(x) = max(|x_1|, ..., |x_n|) = ||x||_\infty$ is the composition of an convex function with non decreasing in all coordinates ($h(y) = max(y_1, ..., y_n)$)) with convex $g_i(x_i) = |x_i|$, meaning that it is convex.

9.7 Jensen's inequality

Assume that f is convex in C. Also assume that X is a discrete random variable that takes values in C, with an arbitrary probability mass function $p_i(x_i)$. Obviously $p_i(x_i) \geq 0$ and $\sum_i p_i(x_i) = 1$. This means

$$\mathbb{E}(X) = \sum_{i=1}^{n} p_i(x_i)x_i$$

is a convex combination of points of C, (belongs to the convex hull of $\{x_1, ..., x_n\}$). meaning that

$$f(\mathbb{E}(X)) = f(\sum_{i=1}^{n} p_i(x_i)x_i) \leq \sum_{i=1}^{n} p_i(x_i)f(x_i) = \mathbb{E}(f(X)).$$

This result can be extended to continuous random variables with an arbitrary density probability function $p_X(x)$, meaning that this inequality holds in general. This leads to the **Jensen's inequality**:

$$f(\mathbb{E}(X)) \leq \mathbb{E}(f(X)),$$

for convex f.

10 Convex optimization problems (COP) in standard form

A convex optimization problem in standard form is an optimization problem of the form:

$$\begin{aligned} \min_{x} \quad & f_0(x) \\ s.t. \quad & f_i(x) \leq 0; \quad i = 1, ..., m \\ & Ax - b = 0; \quad A \in \mathbf{R}^{p \times n} \end{aligned}$$

with:

- $x \in \mathbf{R}^n$ is the *optimization variable*,
- $f_0 : \mathbf{R}^n \to \mathbf{R}$ is convex (*objective function*),
- $f_i : \mathbf{R}^n \to \mathbf{R}$ are convex (*inequality constraints functions*),
- $Ax - b$, $A \in \mathbf{R}^{p \times n}$ are the *equality constraints functions*.

The **domain** D of the optimization problem is $D = \bigcap_{i=0}^{m} Dom(f_i)$.

A point x is **feasible** if belongs to D and fulfils all the constraints.

F is the **feasible set**, that contains all the feasible points.

If $F = \emptyset$, we say that the problem is **unfeasible**.

A point $x^* \in F$ is a solution for the optimization problem when $\forall x \in F, f_o(x^*) \leq f_o(x)$. The **optimal value** p^* is defined as $f_0(x^*) = p^*$.

A problem is **unbounded below** when the function f_0 is unbounded below in F. For instance $f_0(x) = x;\ x \in R$. In this case we define $p^* = -\infty$.

If the problem is unfeasible, we define $p^* = +\infty$.

11 Conditions of minimum for differentiable convex functions

11.1 For once differentiable functions, $\nabla f(x^*) = 0$ is a sufficient condition for a minimum in x^*

but obviously it is not necessary.

11.2 For once differentiable functions, $\nabla f(x^*)^T(y - x^*) \geq 0, \forall y \in C$ is a necessary and sufficient condition for a minimum in x^*

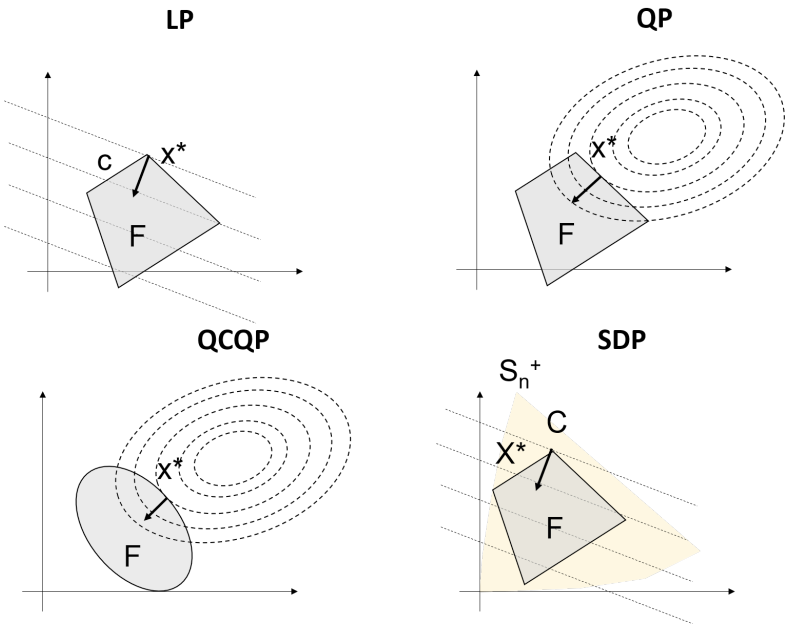
If $x*$ is an optimal point with $\nabla f(x^*) \neq 0$, this means that x^* must be the tangent point of a supporting hyperplane of C. More specifically, this supporting hyperplane has equation:

$$\nabla f(x^*)(x - f(x^*)) = 0.$$

Later we will see a more useful criteria, given by the so called **KKT conditions**.

12 Important examples of convex optimization problems

We will see now several important optimization problems that appear frequently in the applications. We start with Linear Programs (LP), Quadratic Programs (QP) and Quadratic Constrained Quadratic Programs (QCQP)



12.1 Linear Programming (LP)

A Linear convex optimization problem, also known as Linear Program (LPs) is an optimization problem for which the objective function is affine (linear) and the feasible set is a polyhedron.

In general they have the form:

$$\begin{aligned} &min_x \quad c^T x + d \\ &s.t. \;\; Ax \leq b \\ &\qquad Gx = d. \end{aligned}$$

LPs can be expressed in **standard form** as ($A \in \mathbf{R}^{p \times n},\ x, c \in \mathbf{R}^n,\ c \in \mathbf{R}^p$):

$$\begin{aligned} &min_x \quad c^T x \\ &s.t. \;\; Ax = b; \\ &\qquad x \geq 0. \end{aligned}$$

This problem has associated a LP in **inequality form** ($y \in \mathbf{R}^p$):

$$\begin{aligned} &max_y \quad b^T y \\ &s.t. \;\; A^T y \leq c. \end{aligned}$$

Note that for feasible x and y we have:

$$y^T A \leq c^T \implies x^T c \geq x^T A^T y \text{ (as } x \geq 0) \implies c^T x \geq b^T y\ .$$

meaning that if the two problems are solvable (i.e. we have feasible optimal points), the solution to the inequality form LP is a lower bound of the solution for the standard form LP (i.e. $c^T x^* \geq b^T y^*$), a result known as **weak duality**.

We will see later that if one of the problems is solvable, it implies that both problems are solvable and $(x^*)^T c = b^T y^*$, a property known as (**strong duality**). In the terminology we will use when studying duality, we will say that both LPs are duals.

We can transform a general LP into a standard form LP by defining a positive vector s such as $Ax + s = b$ (**slack variables**), and two positive vectors x^+ and x^-, such that $x = x^+ - x^-$. Then, the LP in general form can be written as:

$$\begin{aligned} &min_x \quad c^T x^+ - c^T x^- + d \\ &s.t. \;\; Ax^+ - Ax^- + s = b \\ &\qquad Gx^+ - Gx^- = d \\ &\qquad x^+ \geq 0,\ x^- \geq 0,\ s \geq 0, \end{aligned}$$

which can be written in standard form.

If the LP has an optimal solution (i.e. it is feasible and not unbounded below) it must be in one of the vertex of the polyhedron. However, an exhaustive search is usually unfeasible as the number of vertex becomes very large, even for moderate-sized problems. The **simplex method** is a popular method for solving LPs, but we will not discuss this in this course. Other methods, such as **interior point methods** which we will discuss in this course, can also be used.

12.2 Quadratic Programming (QP)

Assume $P \in S_n^+$ (otherwise use $\frac{1}{2}(P + P^T)$):

$$\begin{aligned} min_x \quad & \frac{1}{2}x^T Px + q^T x + r \\ s.t. \quad & Gx - h \leq 0; \\ & Ax - b = 0. \end{aligned}$$

12.3 Quadratically Constrained QP (QCQP)

Assume $P \in S_n^+$ and $P_i \in S_n^+$; $i = 1, ..., m$:

$$\begin{aligned} min_x \quad & \frac{1}{2}x^T Px + q^T x + r \\ s.t. \quad & \frac{1}{2}x^T P_i x + q_i^T x + r_i \leq 0 \\ & Ax - b = 0. \end{aligned}$$

12.4 Semi Definite Programming (SDP)

A Semidefinite program (SDP) is a minimization problem of the form ($C, A_i, X \in R^{n \times n}$):

$$\begin{aligned} min_X \quad & <C, X> = \mathrm{Tr}(C^T X) \\ s.t. \quad & <A_i, X> = \mathrm{Tr}(A_i^T X) = b_i; \quad i = 1, ..., p \\ & X \geq 0. \text{ (i.e. } X \in S_{++}) \end{aligned}$$

Associated with this SDP we have a **dual SDP** in the form ($S \in R^{n \times n}$):

$$max_y \quad b^T y$$

$$
\begin{aligned}
s.t. \quad & \sum_{i=1}^{p} y_i A_i + S = C, \\
& S \geq 0. \ \text{(i.e. } S \in S_{++}\text{)}
\end{aligned}
$$

We can easily check **weak duality**:

$$\mathrm{Tr}(C^T X) = \sum_{i=1}^{p} y_i \mathrm{Tr}(A_i^T X) + \mathrm{Tr}(S^T X) = b^T y + \mathrm{Tr}(S^T X).$$

But, as X and S are PSD matrices, $\mathrm{Tr}(S^T X) \geq 0$. We use the SVD of $S = U \Lambda U^T = \sum_i \sigma_i u_i u_i^T$ with $\sigma_i \geq 0$:

$$\mathrm{Tr}(S^T X) = \mathrm{Tr}(SX) = \sum \sigma_i \mathrm{Tr}(u_i u_i^T X) = \sum \sigma_i \mathrm{Tr}(u_i^T X u_i) = \sum \sigma_i u_i^T X u_i \geq 0,$$

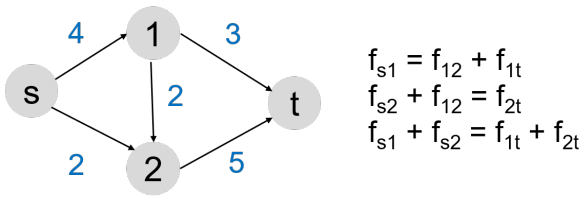
which means that for all feasible X and y we have $\mathrm{Tr}(C^T X) \geq b^T y$. We'll see that in very general conditions (e.g. both feasible sets have non-empty relative interior) **strong duality** also holds.

13 Application examples

13.1 Max flow, min cut

13.1.1 The max-flow problem

Assume that we have a weighted directed graph, with vertices (nodes) $V = \{s, t, u, v, w, ...\}$ and edges (links) $E = \{(s, u), (v, w), ...\}$. s is the source and t is the sink (we assume no edges entering the source of leaving the sink). The links $(u., v)$ has a positive maximum capacity $b_{u,v}$, and are assigned a positive flow $f_{u,v}$, which must be lower than the link capacity. Moreover, for each node v (except for nodes s and t) we must have *flow conservation*: $\sum_{(u,v),\ u \in V} f_{u,v} = \sum_{(v,w),\ w \in V} f_{v,w}$ (recall *Kirchoff law* in electrical circuits). We also have that the flows leaving the source node must be the same as the flows entering the sink node: $\sum_{(s,u),\ u \in V} f_{s,u} = \sum_{(w,t),\ w \in V} f_{w,t}$.



The **max-flow** problem tries to maximize the net flow leaving the source node and can be formulated as an LP:

$$
\begin{aligned}
&max_{f_{uv}} \quad \sum_{(s,u)\in E,\ u\in V} f_{s,u} \\
&s.t. \quad \sum_{(u,v)\in E,\ u\in V} f_{u,v} = \sum_{(v,w)\in E,\ w\in V} f_{v,w};\ v \in V - \{s,t\} \\
&\qquad 0 \le f_{uv} \le b_{uv}.
\end{aligned}
$$

For instance, in the example graph, we have:

$$
f = \begin{bmatrix} f_{s1} \\ f_{s2} \\ f_{12} \\ f_{1t} \\ f_{2t} \end{bmatrix},\ b = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},\ A = \begin{bmatrix} 1 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \end{bmatrix},\ c = \begin{bmatrix} 4 \\ 2 \\ 2 \\ 3 \\ 5 \end{bmatrix}.
$$

and we obtain the LP:

$$
\begin{aligned}
&max_f \quad b^T f \\
&s.t.\ \ Af = 0 \\
&\qquad f \ge 0 \\
&\qquad f \le c.
\end{aligned}
$$

Using s as a vector of slack variables, we get the max-flow expressed as an LP in standard dual form:

$$
\begin{aligned}
&min_{f,s} \quad \begin{bmatrix} -b^T & 0 \end{bmatrix} \begin{bmatrix} f \\ s \end{bmatrix} \\
&s.t. \quad \begin{bmatrix} A & 0 \\ I & I \end{bmatrix} \begin{bmatrix} f \\ s \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix}, \\
&\qquad \begin{bmatrix} f \\ s \end{bmatrix} \ge 0.
\end{aligned}
$$

13.1.2 Optional: the min-cut problem

The **min-cut problem** is formulated as follows: We define a **cut** as a set of vertex of the graph that contains s and do not contain t: $E = \{u \in V |\ s \in E,\ t \in E^c\}$. The **capacity of the cut** E is defined as $\sum_{u\in E} \sum_{v\in E^c} c_{u,v}$, which can be expressed as a product $c^T x$, where $e_{u,v}$ is a vector of 0's and 1's (defined only when $(uv) \in E$. For a given cut S we will have $x_{u,v} = 1$ if $u \in S$, and $v \in S^c$. We will have $x_{uv} = 0$ otherwise. For instance, in our example graph, assume $S = \{s, 1\}$, then we have: $e^T = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \end{bmatrix}$, meaning that $c^T e = 4$.

We can express this problem as LP introducing the variables z_u defined for $u \neq s,t$ as $z_u = 1$ if $u \in S$ and $z_v = 0$ if $v \in S^c$. We have the following inequalities:

$$d_{uv} \geq 0;\ (u,v) \in E,$$

$$d_{uv} - z_u + z_v \geq 0;\ (u,v) \in E,\ u \neq s,\ v \neq t$$

that ensures that if $z_u = 1$ and $z_v = 0$, $d_u v \geq 1$.

$$d_{sv} + z_v \geq 1;\ (s,v) \in E,\ v \neq t,$$

that ensures that we only count d_{sv} if $z_v = 0$.

$$d_{ut} - z_u \geq 0;\ (u,t) \in E,\ u \neq s,$$

that ensures that $d_{uv} \geq 1$ if $z_u = 1$.

$$d_{st} \geq 1, \text{if } (s,t) \in E,$$

In our example, we have four possible cuts:

$$S_1 = \{s\},\ S_2 = \{s,1\},\ S_3 = \{s,2\},\ S_4 = \{s,1,2\}$$

and vectors:

$$d = \begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix},\ z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix},\ z_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},\ z_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},\ z_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\ z_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

That lead to the inequalities:

$$\begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix} + \begin{bmatrix} +z1 \\ +z2 \\ -z1 + z2 \\ -z1 \\ -z2 \end{bmatrix} \geq \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix};\quad \begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix} \geq \mathbf{0},$$

which can be expressed as:

$$\begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} z1 \\ z2 \end{bmatrix} \geq \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

For instance, for the cut S_1, with $z_1 = z_2 = 0$ we have:

$$\begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix} \geq \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

meaning that the solution that minimizes $c^t d$ is:

$$\begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

that correctly finds that only the capacities through the edges $(s, 1)$ and $(s, 2)$ must be counted in this cut for find the cut capacity.

For instance, for cut S_2 we would have:

$$\begin{bmatrix} d_{s1} \\ d_{s2} \\ d_{12} \\ d_{1t} \\ d_{2t} \end{bmatrix} \geq \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

which correctly recognizes that the capacities in $(s, 2), (1, 2)$ and $(1, t)$ must be counted to compute the cut capacity.

We call now $-d = y$, and $-z = w$, obtaining:

$$y + A^T w \leq (-b)$$

$$y \leq 0.$$

This can be established to be true in general, meaning that we can write the min-cut problem as

$$max_{y,w} \quad \begin{bmatrix} 0 & -c^t \end{bmatrix} \begin{bmatrix} w \\ y \end{bmatrix}$$

$$s.t. \quad \begin{bmatrix} A^T & I \\ 0 & I \end{bmatrix} \begin{bmatrix} w \\ y \end{bmatrix} \leq \begin{bmatrix} -b \\ 0 \end{bmatrix}.$$

Finally ,we can see that the max-flow and the min-cut problems can be formulated as dual LPs. Under the conditions of strong duality, **the max-flow is the same as the min-cut**.

13.2 Least squares, Ridge regression and Lasso

Least squares regression is can be formulated as an unconstrained convex optimization problem:

$$min_x \quad ||Ax + b||_2^2.$$

In the notation we used in SANS, the vector $b = -y$, the negative of the measured responses, $A = \Phi(x)$, the matrix with the measured input parameters for our model (that we called x), while $x = \theta$, the coefficients of our regression, meaning that we would write this equation as:

$$min_\theta \quad ||y - \Phi(x)\theta||_2^2.$$

Note that $||Ax+b||_2^2 = (Ax+b)^T(Ax+b) = x^TA^TAx+2b^TAx+b^Tb$, and A^TA is a positive definite matrix if columns of A are independent, meaning that the graph of $||Ax + b||_2^2$ is a paraboloid.

When A is a columns full rank matrix (i.e. all the columns are independent vectors) the problem has an explicit solution given by the **normal equations**:

$$x^* = -(A^TA)^{-1}A^Tb.$$

We can add constraints on the values of the coefficients as a form of regularization. For instance, for **ridge regression** in **constrained form**:

$$min_x \quad ||Ax + b||_2^2$$

$$s.t. \quad ||x||_2^2 \leq R,$$

which can be equivalently formulated in **regularization form** as:

$$min_x \quad ||Ax + b||_2^2 + \lambda||x||_2^2,$$

with $\lambda \geq 0$.

The graph of $\lambda||x||_2^2$ is obviously another paraboloid, and the sum of two paraboloids is another paraboloid: $x^T(A^TA + \lambda I)x + 2b^TAx + b^Tb = ((A + \lambda I)x + b)^T((A + \lambda I)x + b) = ||(A + \lambda I)x + b||_2^2$, meaning that the ridge regression case also has a closed form solution.

Finally, we can think on **Lasso regression**, which promotes sparse solutions. In its constrained form we have:

$$min_x \quad ||Ax + b||_2^2$$

$$s.t. \quad ||x||_1 \leq R,$$

while in regularization form we have:

$$min_x \quad ||Ax + b||_2^2 + \lambda||x||_1.$$

13.3 Compressed sensing

Assume that e have a vector x of M components. We know that most entries of this vector are zero (i.e. it is an sparse vector), but we do not know where they are located (and we only now an upper bound of how many components are non-zero, although this can be relaxed).

We observe the signal x by using n sensors, placed at S different locations, whose readings are a vector b.

We model the sensors reading as $Ax = b$ where A is a given and known matrix (we assume here perfect readings, i.e. no measuring errors, although this can be relaxed).

We wonder whether it is possible to recover the signal when the number of sensors $n << M$, although n should be at least as large as S the **sparsity** of the signal, i.e. the number of non-zero components.

This problem is known as **compressed sensing** or **sparse recovery**.

We could think in solving the problem through the optimization program:

$$min_x \quad \text{card}(x) = ||x||_0$$

$$s.t. \;\; Ax = b.$$

The *cardinality* function $\text{card}(x)$ (also written as $||x||_0$) gives the number of non-zero components of x, and it is non convex.

We can solve this problem by trying with all possible equations with 1 component of z non-zero, then with 2 non-zero components, etc until we reach a solvable system. This however becomes exponentialy expensive when M grows (For instance, if $n \approx M$ we will arrive to the solution after checking around 2^M linear systems of equations).

A possible approach is to solve instead the convex program:

$$min_x \quad ||x||_1$$

$$s.t. \;\; Ax = b,$$

which is convex. In fact this can be transformed into a LP introducing the auxiliary variables w:

$$min_{x,w} \quad w_1 + ... + w_M$$

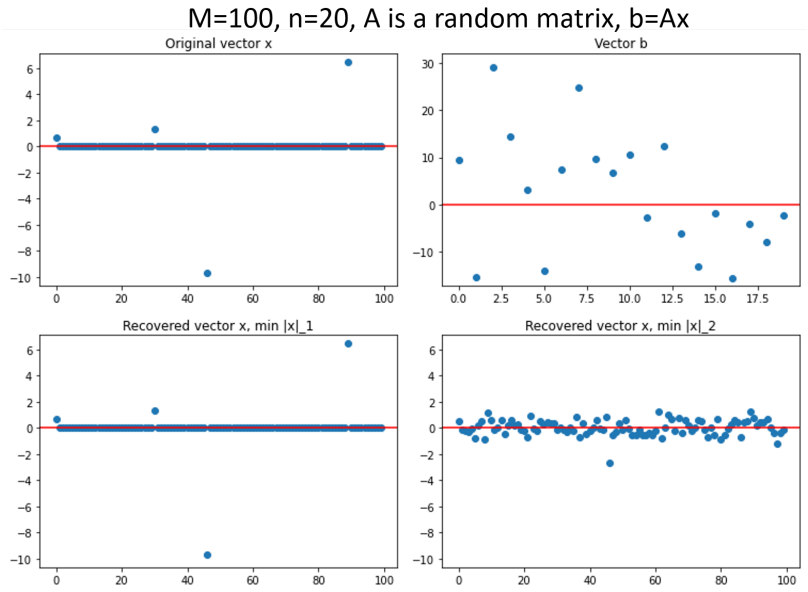
$$s.t. \;\; Ax = b.$$

$$x_i - w_i \leq 0$$

$$-x_i - w_i \leq 0.$$

As an example, assume $M = 100$ and $n = 20$, for a signal of sparsity 4. We use a random matrix A. We plot the vector x and the recovered signal using the $||x||_1$

norm. We also plot what would be the solution if we use the $||x||_2$ norm instead. In this case we have an explicit solution $x^* = A^\dagger b$ where $A^\dagger = A^T(AA^T)^{-1}$ is the pseudoinverse of A, which is a fat and short matrix (i.e. it is full row rank, not full column rank as we have in the least square solution of linear regression):



Sometimes though, the recovered signal is incorrect. **Compressed sensing** studies under which assumptions we can expect the recovered signal to be correct.

13.4 Sparse correlation graph (Graphical lasso)

Assume that we have n sensors that give measures $x = (x_1, .., x_n)$ generated by a multivariate gaussian random variable X. We want to determine the inverse correlation matrix $\Lambda = \Sigma^{-1}$ (i.e. the *precission* matrix). We are interested in a sparse estimation.

The elements of Σ are the covariances between the different components of the vector:

$$\Sigma_{ij} = \sigma_{ij} = \mathbb{E}((x_i - \mu_i)^T(x_j - \mu_j)).$$

The interpretation of Λ is a bit more complicated. An essential fact is that $\Lambda_{ij} = 0$ iff the random variables X_i and X_j are conditionally independent given the other components of the random vector X.

Assume that we have p samples of X: $\{x^{(1)}, ..., x^{(p)}\}$ (each sample is a vector of n components). We want to estimate μ and Λ. We can easily see that the ML estimator of μ is:

$$\hat{\mu} = \frac{\sum_{i=1}^{p} x^{(i)}}{p}.$$

To find the ML estimator of Λ we must maximize:

$$\frac{(det(\Lambda)^{1/2})^p}{(2\pi)^{\frac{np}{2}}} e^{-\sum_{i=1}^{p}(x^{(i)}-\hat{\mu})^T\Lambda(x^{(i)}-\hat{\mu})}.$$

This is a non-convex function. However, if take the log of the ML function, we obtain a concave function (i.e. this is a **log-concave** function), and our optimization problem is equivalent to minimizing the minus log-likelihood (divided by p):

$$-log(det(\Lambda)) + \frac{1}{p}\sum_{i=1}^{p}(x^{(i)} - \hat{\mu})^T\Lambda(x^{(i)} - \hat{\mu}).$$

We can give another interpretation to the formula using $tr(ABC) = tr(CBA)$:

$$\sum_{i=1}^{p}(x^{(i)}-\hat{\mu})^T\Lambda(x^{(i)}-\hat{\mu}) = tr(\sum_{i=1}^{p}(x^{(i)}-\hat{\mu})^T\Lambda(x^{(i)}-\hat{\mu})) = tr(\sum_{i=1}^{p}(\Lambda x^{(i)}-\hat{\mu})(x^{(i)}-\hat{\mu})^T).$$

We recognize here $S = \frac{1}{p}\sum_{i=1}^{p}(x^{(i)}-\hat{\mu})(x^{(i)}-\hat{\mu})^T$ the sample variance-covariance matrix, leading to the following COP:

$$min_x \quad -log(det(\Lambda)) + tr(\Lambda S)$$

$$s.t. \ \Lambda \in S_n^{++}.$$

We can use the $||.||_1$ regularization to obtain an sparse solution:

$$min_x \quad -log(det(\Lambda)) + tr(\Lambda S) + \lambda||\Lambda||_1.$$

$$s.t. \ \Lambda \in S_n^{++},$$

where the 1-norm is applied to all the elements of Λ as it was a vector.

Note that the three terms of the objective function are convex. $tr(A^TB)$ can be interpreted as the scalar product between A and B, while $\Lambda \in S_n^{++}$ is a convex set.

13.5 Optimal sensor location

We consider a signal x, a vector of length S. We use $M \geq S$ sensors for monitoring x. These sensors obtain a reading y given by $y = Ax$ for some known matrix A, with rows ϕ_i^T.

We assume now that Gaussian noise in the readings, that we model as M; i.i.d Gaussian r.v. ϵ_i with $\mathbb{E}(\epsilon_i) = 0$ and variance $\sigma_i^2 = \sigma^2$, meaning that our readings are in fact:

$$y = Ax + \epsilon.$$

The maximum likelihood estimation for x is:

$$\hat{x} = (A^T A)^{-1} A^T y.$$

This estimator is unbiased:

$$\mathbb{E}(\hat{X}) = (A^T A)^{-1} A^T \mathbb{E}(Y) = (A^T A)^{-1} A^T \mathbb{E}(AX + \epsilon) = \mathbb{E}(X).$$

The variance of this estimator is:

$$\mathbb{E}((\hat{X} - X)(\hat{X} - X)^T) = \mathbb{E}(\hat{X}\hat{X}^T) - \mathbb{E}(\hat{X}X^T) - \mathbb{E}(X\hat{X}^T) + \mathbb{E}(XX^T).$$

These terms can be simplified, for instance:

$$\mathbb{E}(\hat{X}\hat{X}^T) = (A^T A)^{-1} A^T \mathbb{E}(YY^t) A (A^T A)^{-1},$$

and (note $\mathbb{E}(E) = 0$ and $\mathbb{E}(EE^T) = \sigma^2 I$):

$$\mathbb{E}(YY^t) = A\,\mathbb{E}(XX^t)A^T + \sigma^2 I,$$

meaning

$$\mathbb{E}(\hat{X}\hat{X}^T) = \mathbb{E}(XX^t) + \sigma^2 (A^T A)^{-1},$$

while:

$$\mathbb{E}(\hat{X}X^T) = \mathbb{E}(XX^T).$$

Finally we get:

$$\mathbb{E}(XX^T) - \mathbb{E}(XX^T) - \mathbb{E}(XX^T) + \mathbb{E}(XX^T) + \sigma^2 (A^T A)^{-1} = \sigma^2 (A^T A)^{-1}.$$

Our problem consists now in selecting $n \leq M$ sensors (with $n \geq S$). This selection must be done in such a way that we minimize the variance of our estimator. We can use a selection matrix C of 1's and 0's which selects the rows of A corresponding to the selected vectors. For instance, assume $S = 3$, $M = 5$, and $n = 3$. If we want to select the readings of sensors at positions 2, 3, and 5 we will use:

$$A' = CA = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & -4 \\ 0 & 2 & 1 \\ -1 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & -4 \\ 0 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}.$$

Now, we get a new matrix $A' = CA$. We can easily see that $(A')^T A' = (A^T H A)$ where H is a diagonal matrix $H = diag(h)$ where the components of vector $h = (h_1, ..., h_M)$ indicate whether we use or not a sensor at a given position (obviously $\mathbf{1}^T z = n$). For instance, in our example:

$$H = C^T C = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

We call ϕ_i^T the row vectors that constitute the rows of matrix A. Under this conditions $(A')^T A' = \sum_i h_i \phi_i \phi_i^T$, and the variance of our estimator after the selection would be:

$$\sigma^2 (\sum_i h_i \phi_i \phi_i^T)^{-1} = \sigma^2 (\begin{bmatrix} -1 & 1 & -4 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ -4 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix})^{-1}.$$

One possible solution for the problem is to find the selection that minimizes the volume of the confidence ellipsoid of the estimation error, that leads to the non convex combinatorial optimization problem:

$$\begin{aligned} & min_x \quad - log\ det(\sum_i z_i \phi_i \phi_i^T), \\ & s.t. \ \ h_i \in \{0, 1\}, \\ & \quad \mathbf{1}^T h = n. \end{aligned}$$

that we must solve by an exahustive search. This is a problem that cannot be solved even for moderately large problems. We can give an approximate solution by solving the (convex) optimization problem:

$$\begin{aligned} & min_x \quad - log\ det(\sum_i h_i \phi_i \phi_i^T), \\ & s.t. \ \ 0 \leq h_i \leq 1, \\ & \quad \mathbf{1}^T h = n, \end{aligned}$$

chosing (for instance) the q coordinates with largest values of h as the selected sensors.

14 Duality

14.1 The Lagrangian function $L(x, \lambda, \mu)$

Assume that we have the (*in general non convex*) optimization problem in the form:

$$
\begin{aligned}
& min_x \quad f_0(x) \\
& s.t. \;\; f_i(x) \leq 0; \quad i = 1, ..., p \\
& \qquad h_i(x) = 0; \quad i = 1, ..., m.
\end{aligned}
$$

We define the **Lagrangian function** $L(x, \lambda, \mu)$ associated to the previous optimization problem as:

$$L(x, \lambda, \mu) = f_0(x) + \sum_{i=1}^{p} \lambda_i f_i(x) + \sum_{i=1}^{m} \mu_i h_i(x),$$

with $x \in D$, $\lambda_i \geq 0$**, and no restrictions on** μ_i. λ_i and μ_i are called **Lagrange multipliers** or **dual variables**.

14.1.1 A simple example of Lagrangian function

Assume that we have the following simple COP:

$$
\begin{aligned}
& min_x \quad \frac{1}{4}x^2 - x \\
& s.t. \;\; x \leq 0.
\end{aligned}
$$

$D = \mathbf{R}$, $F = \mathbf{R}_-$, the optimal point is $x^* = 0$, and the optimal value is $p^* = 0$.

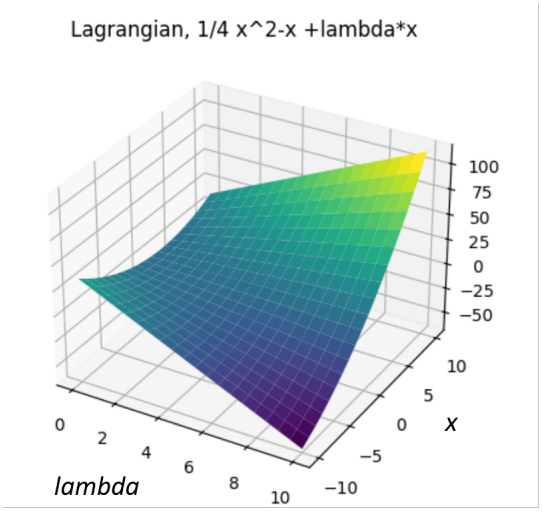
The Lagrangian function associated with this problem is:

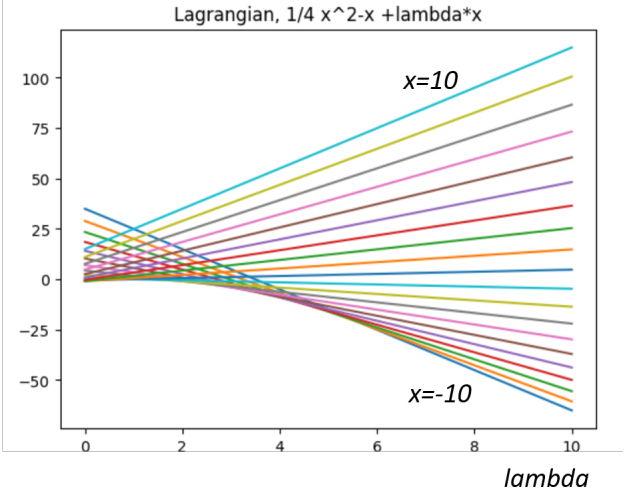
$$L(x, \lambda) = \frac{1}{4}x^2 - x + \lambda x, \;\; x \in \mathbf{R}, \; \lambda > 0.$$

14.1.2 Graphical representations of the Lagrangian

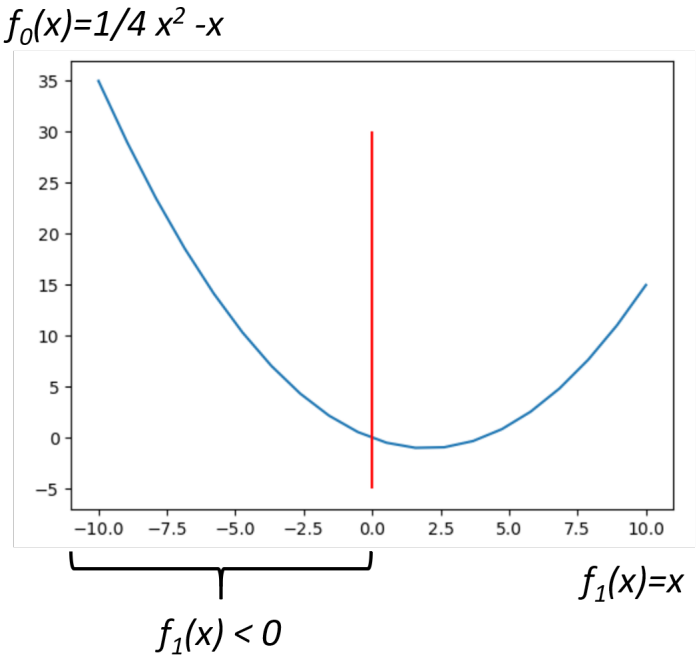
In this example, we can see a graphical representation of the Lagrangian function:

$$L(x, \lambda) = \frac{1}{4}x^2 - x + \lambda x, \;\; x \in \mathbf{R}, \; \lambda > 0.$$





An alternative representation is given by this plot:



14.1.3 The terms $\sum_{i=1}^{p} \lambda_i f_i(x) + \sum_{i=1}^{m} \mu_i h_i(x)$ are an approximation to the indicator functions $I_F(x)$

This looks as a very bad approximation, but as we will see, this approximation is surprisingly useful.

In general, we have:

$$\lambda \geq 0,\ \sum_{i=1}^{p} \lambda_i f_i(x) + \sum_{i=1}^{m} \mu_i h_i(x) \leq I_F(x),$$

and

$$sup_{\lambda \geq 0, \mu}\ (\sum_{i=1}^{p} \lambda_i f_i(x) + \sum_{i=1}^{m} \mu_i h_i(x)) = I_F(x).$$

In our example, $F = \{x; x \leq 0\} = \mathbf{R}_-$, and we can define the following equivalent unconstrained OP:

$$min_x\ \frac{1}{4}x^2 - x + I_{\mathbf{R}_-}(x).$$

14.1.4 In F, the Lagrangian is always a lower bound of $f_0(x)$.

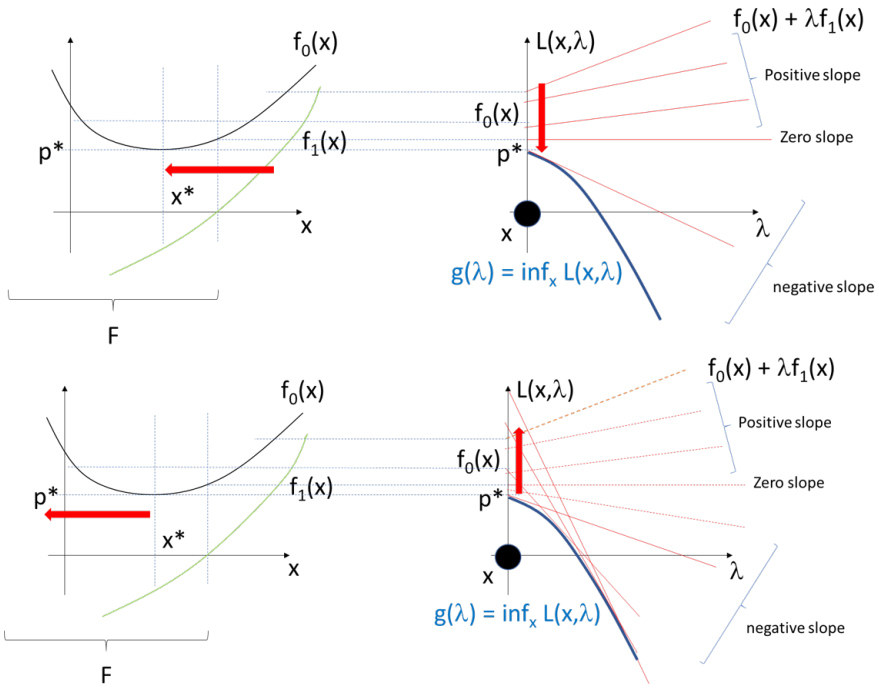
In the feasible set, $\lambda_i f_i(x) \leq 0$, while $\mu_i h_i(x) = 0$, meaning that:

$$L(x, \lambda, \mu) \leq f_0(x) + I_F(x) = f_0(x); \quad x \in F.$$

14.2 The Lagrangian dual function $g(\lambda, \mu)$

For a Lagrangian function $L(x, \lambda, \mu)$ we define the **Lagrangian dual function**:

$$g(\lambda, \mu) = inf_{x \in D}\ L(x, \lambda, \mu).$$



14.2.1 The Lagrange dual function $g(\lambda, \mu)$ is always concave

For a fixed x, the Lagrangian function has the form

$$a + \sum_i b_i \lambda_i + \sum_i c_i \mu_i; \quad a = f_0(x); \; b_i = f_i(x); \; c_i = h_i(x).$$

In other words, it is an *affine function of* λ *and* μ, which is means the it is both convex and concave.

The Lagrangian dual is the pointwise infimum of concave functions, which means that: **The function $g(\lambda, \mu)$ is a concave function, regardless of whether the functions $f_i(x)$ and $h_i(x)$ are convex or not**.

14.2.2 The Lagrangian dual function is a lower bound of p^*, i.e. $g(\lambda, \mu) \leq p*$

Note that $(F \subset D)$:

$$g(\lambda, \mu) = inf_{x \in D} \; (L(x, \lambda, \mu)) \leq inf_{x \in F} \; (L(x, \lambda, \mu)).$$

Moreover, we know that in the feasible set:

$$L(x, \lambda, \mu) \leq f_0(x); \quad x \in F,$$

which means:

$$inf_{x\in F}\ (L(x,\lambda,\mu)) \leq inf_{x\in F}\ f_0(x) = p*\,.$$

Then:

$$g(\lambda,\mu) \leq p*\,.$$

14.2.3 Lagrangian dual functions of equivalent optimization problems may be different

Introducing small transformations in the primal optimization problem to generate equivalent optimization problems, can lead to very different dual problems.

As a simple example, assume the following COP:

$$min_x \quad -x$$

$$s.t.\ \ e^x - 1 \leq 0,$$

with $D = \mathbf{R}$ and $F = (-\infty, 0]$ and $p^* = 0$.

The Lagrangian function would be: $L(x,\lambda) = -x + \lambda(e^x - 1)$, with $\frac{\partial L}{\partial x} = -1 + \lambda e^x = 0 \implies x = \log(\frac{1}{\lambda})$ meaning that $g(\lambda) = \log(\lambda) + 1 - \lambda$. Its maximum is in $\frac{dg}{d\lambda} = \frac{1}{\lambda} - 1 = 0 \implies \lambda^* = 1$, which means $d^* = 0$.

For the equivalent COP (which is now a LP):

$$min_x \quad -x$$

$$s.t.\ \ x \leq 0,$$

the Lagrangian function would be: $L(x,\lambda) = -x + \lambda x = (1-\lambda)x$. The infimum of this function is $-\infty$, except for $\lambda = 1$, which leads to $g(\lambda) = -\infty,\ \lambda \neq 1$, $g(1) = 0$ (still a concave function). Obviously, in this case we have again $d^* = 0$.

Let us consider an equivalent optimization problem, but now in a *non-convex form*:

$$min_x \quad -x$$

$$s.t.\ \ f_1(x) \leq 0,$$

with $f_1(x) = x$ if $x \leq 0$ and $f_1(x) = \frac{1}{2}x$ if $x \geq 0$.

Now $L(x,\lambda) = -x + \lambda f_1(x) = (-1+\lambda)x$ if $x \leq 0$ and $(-1 + \frac{1}{2}\lambda)x$ of $x \geq 0$.

The dual function $g(\lambda)$ will be given by $g(\lambda) = -\infty$ if $0 \leq \lambda < 1$ and $g(\lambda) = -\infty$ if $\frac{1}{2} \leq \lambda$, or in other words $g(\lambda) = -\infty$ for $\lambda \geq 0$, which means $d^* = -\infty$. Observe that in this non-convex problem the duality gap is infinite.

14.3 Examples of Lagrange dual functions

14.3.1 LP in standard form

Assume the following LP in standard primal form ($x, c \in \mathbf{R}^n,\ A \in \mathbf{R}^{p\times n},\ b \in \mathbf{R}^p$):

$$min_x \quad c^T x$$

$$s.t.\ \ Ax - b = 0$$

$$-x \leq 0.$$

The Lagrangian function is ($\lambda \in \mathbf{R}^n,\ \lambda \geq 0\ \mu \in \mathbf{R}^p$):

$$L(x, \lambda, \mu) = c^T x - \lambda^T x + \mu^T (Ax - b) = (c^T - \lambda^T + \mu^T A)x - \mu^T b,$$

The Lagrangian is a linear function, with infimum $-\infty$ if $c^T - \lambda^T + \mu^T A \neq 0$, while the infimum is $-\mu^T b$ if $c^T - \lambda^T + \mu^T A = 0$, i.e.:

$$g(\lambda, \mu) = -\mu^T b;\ \ if\ \ c^T - \lambda^T + \mu^T A = 0,$$

$$g(\lambda, \mu) = -\infty;\ \ if\ \ c^T - \lambda^T + \mu^T A \neq 0.$$

which leads to the following maximization problem:

$$max_{\mu,\lambda} \quad b^T(-\mu)$$

$$s.t.\ \ A(-\mu) = \lambda + c.$$

$$\lambda \geq 0.$$

With $\mu = -y$ and $\lambda \geq 0$ treated as an slack variable, we obtain the dual problem in standard form:

$$max_y \quad b^T y$$

$$s.t.\ \ Ay \geq c.$$

14.3.2 QP: Minimal norm solution for a linear system of equations

Assume that we want to find the minimim euclidian norm solution of $Ax = b$, where A is a full row-rank matrix (i.e. a short fat matrix with more column than rows, while all the rows are independent).

$$min_x \quad ||x||_2^2 = x^T x$$

$$s.t. \qquad Ax - b = 0; \quad A \in \mathbf{R}^{p\times n}.$$

In this case, the solution can be expressed as:

$$x^* = A^\dagger b = A^T (AA^T)^{-1} b.$$

To see that, assume that x is a feasible point, which implies $Ax = b$, which implies $A(x - x^*) = 0$, which implies:

$$(x - x^*)^T x^* = (x - x^*)^T A^T (AA^T)^{-1} b = (A(x - x^*))^T (AA^T)^{-1} b = 0.$$

meaning that $x - x^*$ and x^* are orthogonal, which implies

$$||x||^2 = ||x^* + x - x^*||^2 = ||x^*||^2 + ||x - x^*||^2 \geq ||x^*||^2,$$

implying that x^* is the indeed the optimal solution.

Note that $(A^T A)^{-1} A^T$ is a projection matrix on the column space of A, while $A^T (AA^T)^{-1}$ is a projection matrix on the row space of A.

We can derive the same result by using the Lagrangian function, which is:

$$L(x, \mu) = x^T x + \mu^T (Ax - b),$$

with $x \in \mathbf{R}^n,\ \mu \in \mathbf{R}^p$.

We first obtain:

$$\nabla_x L(x, \mu) = 2x + A^T \mu = 0,$$

with solution $x = -\frac{1}{2} A^T \mu$.

From here we obtain the Lagrange dual function:

$$g(\mu) = \frac{1}{4} \mu^T AA^T \mu - \frac{1}{2} \mu^T AA^T \mu - b^T \mu = -\frac{1}{4} \mu^T AA^T \mu - b^T \mu,$$

whose solution is $\mu^* = -2(AA^T)b$, which leads to

$$x^* = A^T (AA^T)^{-1} b.$$

14.3.3 Using the dual problem to give a lower bound of the solution: the two-way partition problem

We want to classify n points into two sets, by creating an indicator vector $x = (x_1, .., x_n)$ for which $x_i = \{+1, -1\}$. Assume $W_{i,j} \leq 0$ measures a (positive) cost for placing i and j in the same set, while the (negative) cost of placing them in different sets is $-W_{i,j}$, (i.e. $W_{i,j}$ measures how much i and j dislikes each other). We assume $W_{ij} = W_{ji}$.

We want to find the optimal x^* which minimizes the over cost function:

$$\sum_i \sum_j x_i x_j W_{i,j} = x^T W x.$$

This is an example of a *combinatorial optimization* problem, and it can be shown that it is a very hard to solve for large values of N. In fact this is an example of an NP *hard problem*.

The (non-convex) optimization problem can be formulated as:

$$min_x \quad x^T W x$$

$$s.t. \quad x_i^2 = 1.$$

Note that the feasible set F (vertices of an hypercube) is a discrete set, meaning that it is non convex.

The Lagrange dual function is:

$$g(\mu) = inf_x \ (x^T W x + \sum_i \mu_i (x_i^2 - 1)) = inf_x \ (x^T (W + diag(\mu)) x - \mathbf{1}^T \mu).$$

This expression is a quadratic form, meaning that the infimum is $-\infty$ unless $W + diag(\mu) \in S_n^+$:

$$g(\mu) = -\mathbf{1}^T \mu, \quad W + diag(\mu) \in S_n^+,$$

$$g(\mu) = -\infty, \quad W + diag(\mu) \notin S_n^+.$$

Any value μ that fulfills $W + diag(\mu) \in S_n^+$ gives the lower bound $-\mathbf{1}^T \mu$ for the optimal value p^*.

For instance, if we take $\mu = -\lambda_{min}(W)\mathbf{1}$, (minimum eigenvalue of W), the condition of PSD holds (we are translating to the right all eigenvalues of W making them ≥ 0), leading to:

$$p^* \geq g(-\lambda_{min}(W)\mathbf{1}) = n\lambda_{min}(W).$$

The minimum eigenvalue $\lambda_{min}(W)$ can be usually found without much computation effort even for very large values of n (for instance, for sparse matrices W).

14.4 The duality gap and Slater's condition for strong duality

We have seen that $p^* - d^* \geq 0$. We define the **duality gap** as $p^* - d^*$. When $p^* = d^*$, we say that we say that **strong duality** holds.

The conditions that the problem has to fulfill for strong duality to hold are called **constraint qualifications**.

For most COPs, strong duality holds, although not always.

For most non-convex optimization problems strong duality does not hold, although for some problems it holds.

14.4.1 Slater's condition for strong duality

A very useful constrain qualification is the **Slater's condition**:

Assume that the non-affine inequality constraints for a COP are the functions $f_i(x)$. **The COP has strong duality if, for all the non-affine constraint function $f_i(x)$, there is at least a feasible point $\hat{x} \in F$ that fulfills $f_i(\hat{x}) < 0$ (for all i)** and $h_i(\hat{x}) = 0$.

In other words, **the feasible set F has a non-empty relative interior**.

The **relative interior of S** is defined as $\{x \in S | \ \exists \epsilon > 0, B_\epsilon \cap \text{convexhull}(S) \subseteq S$.

14.5 The Karush-Kuhn-Tucker (KKT) conditions

Assume that strong duality holds (i.e. $p^*=d^*$) in a (non-necessarily) convex problem. Then:

$$p^* = f_0(x^*) = d^* = g(\lambda^*, \mu^*) = inf_{x \in D} \ (f_0(x) + \sum_i \lambda_i^* f_i(x) + \sum_i \mu_i^* h_0(x)) \leq$$

$$\leq inf_{x \in F} \ (f_0(x) + \sum_i \lambda_i^* f_i(x) + \sum_i \mu_i^* h_i(x)).$$

But in F, we have $\lambda_i^* f_i(x) \leq 0$ and $\mu_i^* h_i(x) = 0$, which implies:

$$inf_{x \in F} \ (f_0(x) + \sum_i \lambda_i^* f_i(x) + \sum_i \mu_i^* h_i(x)) \leq inf_{x \in F} \ (f_0(x)) = f_0(x^*) = p^*.$$

This means that **the inequalities are in fact equalities**.

14.5.1 Stationarity

From the previous chain of inequalities (that in fact are equalities) we have that the convex differentiable function $h(x)$ defined as

$$h(x) = L(x, \lambda^*, \mu^*),$$

has a minimum at $x = x^*$, meaning that $h'(x^*) = 0$, or in other words: **The gradient with respect to x of the Lagrangian vanishes at optimal primal and dual points**:

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0,$$

a condition known as **stationarity**

14.5.2 Complementary slackness

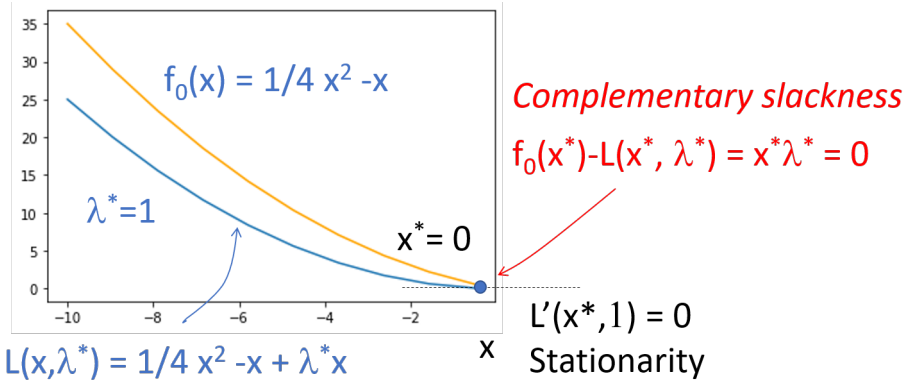
In the feasible set F we have: $L(x^*, \lambda^*, \mu^*) = p^*$, which implies:

$$\sum_i \lambda_i^* f_i(x^*) = 0,$$

which in turn implies:

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, ..., p,$$

a condition called (**complementary slackness**).



14.5.3 Karush-Kuhn-Tucker (KKT) conditions

We already know that, for unconstrained COP with differentiable objective function f_0, the necessary and sufficient condition for optimality is:

$$\nabla f_0(x^*) = 0.$$

Now we want to find equivalent optimality conditions for constrained COP. Let us assume that we have a COP, in which *strong duality holds*, with *differentiable* functions $f_i(x)$, $i = 0, .., p$. The KKT conditions establish **necessary and sufficient conditions** for optimality:

- *Primal feasibility*:

  $$f_i(x^*) \leq 0; \quad i = 1, ..., p;$$

  $$Ax^* - b = 0.$$

- *Dual feasibility*:

  $$\lambda_i^* \geq 0.$$

- *Stationarity*:

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0.$$

- *Complementary slackness*:

$$f_i(x^*)\lambda_i^* = 0 \;\; i = 1, ..., p.$$

If the problem is nonconvex, but still we have strong duality, the KKT conditions are necessary but not sufficient.

14.5.4 Example: KKT conditions for QP problems with equality constraints

In an unconstrained QP problem, of the form ($P \in S_n^{++}$):

$$min_x \quad \frac{1}{2}x^T P x + q^T x.$$

The optimal point can be found by solving the linear equation:

$$Qx^* = -q.$$

We now will find a generalization of this results for QPs with linear equality constraints.

The constrained QP:

$$min_x \quad \frac{1}{2}x^T P x + q^T x$$

$$s.t. \qquad Ax - b = 0; \quad A \in \mathbf{R}^{q \times n},$$

fulfills the Slater's conditions as long that we have a non-empty feasible set F, meaning that strong duality holds.

The Lagrangian function is:

$$L(x, \mu) = \frac{1}{2}x^T P x + q^T x + \mu^T (Ax - b),$$

with $x \in \mathbf{R}^n, \; \mu \in \mathbf{R}^q$.

The KKT conditions would be:

- *Primal feasibility*:

$$Ax^* = b.$$

- *The gradient respect to x of the Lagrangian vanishes at optimal primal and dual points*:

$$Px^* + A^T \mu^* = -q.$$

These equations can be expressed in compressed form as (KKT system):

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \mu^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix} \tag{14.5.1}$$

This will be an important expression when we will study Newton's method.

14.5.5 Example: waterfiling

15 Solution methods

During the last years there have been considerable advances in solution methods for COP. Today, for many applications, there are libraries that find the solution of COP in few seconds even in laptops, although obviously the numerical solution of some very large problems remain challenging and special algorithms and execution in clusters is required.

As a contrast, many non-convex optimization, as the one we find when training large Neural Networks, require high-performance computational resources and long convergence times even to find an approximate solution.

15.1 Unconstrained COP

We want to find a method for solving the unconstrained COP:

$min_x f_0(x)$.

We assume that the function $f_0(x)$ is twice differentiable and the problem is solvable.

As $f_0(x)$ is twice differentiable and convex, a sufficient and necessary condition for an optimal point is:

$\nabla f_0(x^*) = 0$.

Except in some special cases, there is not explicit formula for solving this equation, and usually we will generate a sequence of values $x^{(k)}$ that converges to the solution. A technical detail is that it is assumed that for the first value $x^{(0)} \in D$, with the set $S = \{x \in D; f_0(x) \leq f_0(x^{(0)})\}$ a closed set which is true for instance if $dom f_0 = \mathbf{R}$ aor if $f_0(x) \to +\infty$ as $x \to$ boundary $Dom f_0$.

15.2 Descent methods

The iteration proceeds as:

$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}$,

where $f(x^{(k+1)} < f(x^{(k)})$. The value $t^{(k)}$ is the **step size**.

15.3 Gradient descent

In **Gradient descent** we set the descent value to $\Delta x = -\nabla f(x)$, using an stopping criteria of the form $||\nabla f(x)||_2 < \epsilon$.

The convergence of this method is too slow and seldom is used in practice for solve convex problems.

15.3.1 Exact line search and backtracking line search

For choosing the value $t^{(k)}$, we can find, at each step, the value of t than minimizes $f_0(x + t\nabla f(x))$, a method called **exact line search**.

An alternative is to iteratively find an optimal step size. The **backtracking line search** algorithm starts with $t^{(0)} = 1$. Then for two values $\alpha \in (0, 1/2)$ and $\beta \in (0, 1)$ the iteration proceeds by $t^{(n+1)} = \beta t^{(n)}$ until

$$f(x - t\nabla f(x)) \leq f(x) - \alpha t||\nabla f(x)||^2.$$

15.4 Steepest descent

A generalization of gradient descent is to use $\Delta x = argmin\{\nabla f(x)^T v|||v|| = 1\}$ for some norm $||.||$.

For instance, gradient descent provides the steepest descent for the euclidean norm.

For instance, for $P \in S_n^{++}$ we can define a *quadratic norm* as $||x||_P = x^T P x$, and in this case the steepest descent will be $\Delta x = -P^{-1}\nabla f(x)$.

In general, steepest descent has convergence properties similar to gradient descent.

15.5 Newton's method

A particular choice for steepest decent is to use $P = H_f(x)$ (i.e. the **Hessian nom**. The corresponding method is known as **Newton method**. As we will see, the convergence properties are completely different in this case.

The **Newton step**, $\Delta_N x$, is given by:

$$\Delta_N x = -H_f(x)^{-1}\nabla f(x).$$

The Newton step can be found by solving a linear equation:

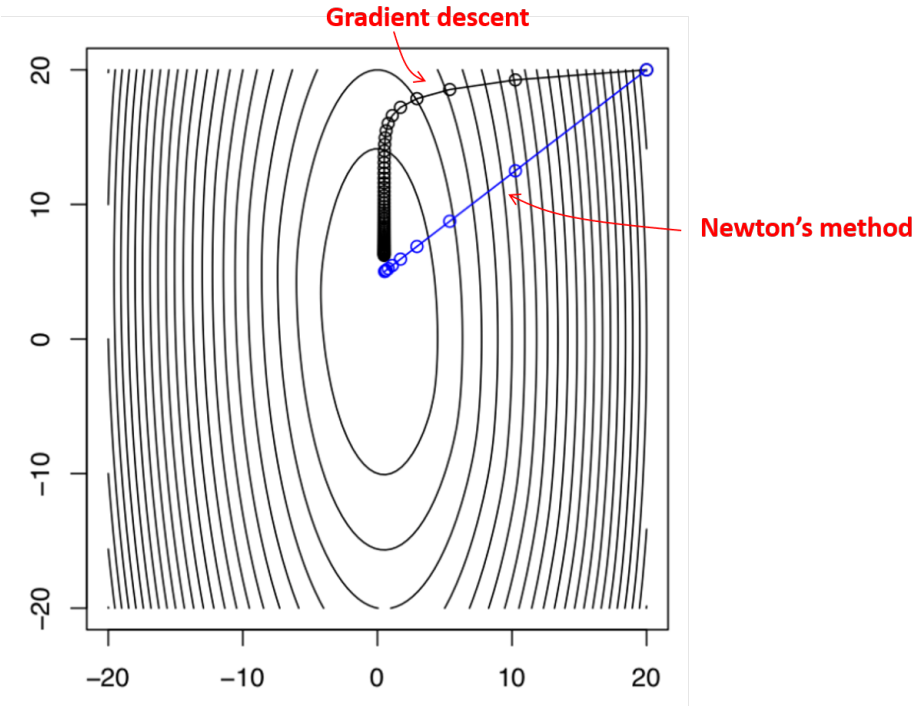
$$H_f(x)\Delta_N x = -\nabla f(x).$$

An interpretation for Newton's method comes from considering the second-order approximation of $f_0(x)$ across the point $x^{(n)}$ of the n-th step of our iteration:

$$f_0(x) = f_0(x^{(n)}) + \nabla f(x^{(n)})^T (x - x^{(n)}) + \frac{1}{2}(x - x^{(n)})^T H_f(x^{(n)})(x - x^{(n)}) + O(||x - x^{(n)}||^2).$$

By defining $\Delta_N x = x^{(n+1)} - x^{(n)}$, we minimize the quadratic form to obtain:

$$H_f(x^{(n)})\Delta_N x = -\nabla f(x^{(n)}).$$



An interesting property of Newton method is that it is invariant to affine transformation of the variables, or in other words is **scale invariant**: Use $x = Ay$, with a full rank square A, and $g(y) = f(Ax)$ then

$$\nabla g(y) = A^T \nabla f(Ay),$$

$$H_g(y) = A^T H_f(Ay) A,$$

and then:

$$y^+ = y - H_g(y)^{-1} \nabla g(y) =$$

$$= y - (A^T H_f(Ay) A)^{-1} A^T \nabla f(Ax) =$$

$$= y - A^{-1}H_f(Ay)^{-1}\nabla f(Ay),$$

meaning

$$Ay^+ = Ay - H_f(Ay)^{-1}\nabla f(Ay),$$

which is the same as

$$x^+ = x - H_f(x)^{-1}\nabla f(x).$$

We define the **Newton decrement** $\lambda(x)$ as:

$$\lambda(x) = (\nabla f(x)^T H_f(x)^{-1}\nabla f(x))^{1/2}.$$

The interpretation of the Newton decrement is the following: The difference between $f_0(x)$ and the value of the minimum of its quadratic approximation (i.e. at $x^* = x - H_f(x)^{-1}\nabla f(x)$) is:

$$f(x) - (f(x) - \nabla f(x)^T H_f(x)^{-1}\nabla f(x) + \frac{1}{2}\nabla f(x)^T H_f(x)^{-1} H_f(x) H_f(x)^{-1}\nabla f(x)) =$$

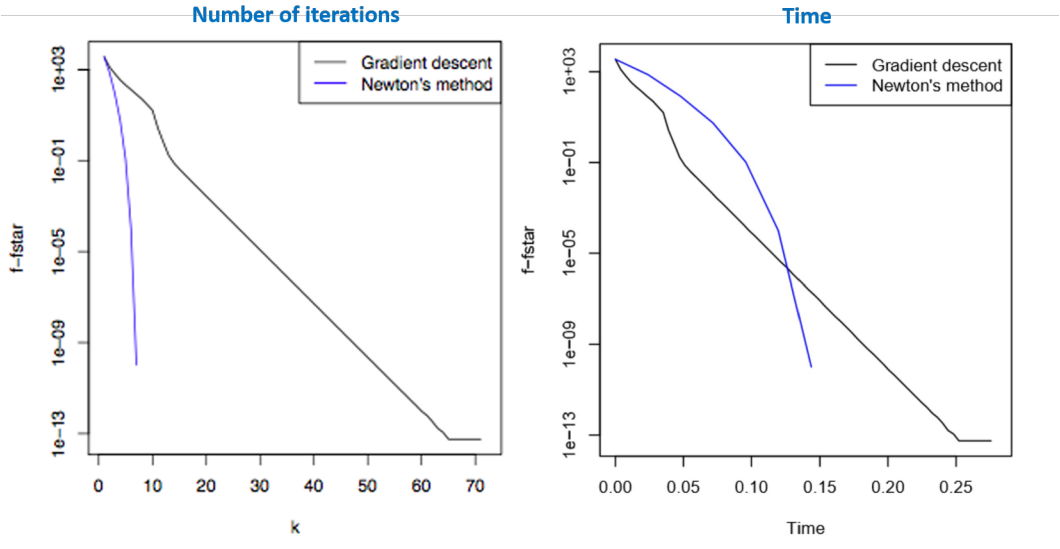
$$= \frac{1}{2}\nabla f(x)^T H_f(x)^{-1}\nabla f(x) = \frac{1}{2}\lambda^2(x),$$

this means that $\frac{1}{2}\lambda^2(x)$ is an approximate upper bound of the optimality gap $f(x) - p^*$.

15.6 Converge properties of Newton's method: damped and quadratic phases

If we plot $\log(f_0(x^{(n)} - p^*)$ versus n (i.e. the iteration number) for gradient descent we usually obtain a **linear dependence**. Depending on the type of problem the slope of this linear dependence can be low, meaning that it can have very poor convergence properties.

On the other hand, Newton's method usually exhibits extremely good convergence properties. If we plot $\log(f_0(x^{(n)} - p^*)$ versus n, we can usually differentiated two regions. Initially we can have a linear dependence (**damped** region), followed by a **quadratic dependence region**. Once we enter into the quadratic regime, we usually reduce the error to float arithmetic error in few iterations, for instance 4 or 5 iterations.



An explanation for this behavior is that, when we are close to the solution, the second-order approximation becomes a very good approximation of the function, and then the Newton's method gives in each step a value very close to the optimal solution.

15.6.1 Convergence analysis for self-concordant functions

This observed behaviour has been proved by Nesterov and Nemirovskii for **self-corcondant** functions.

A **self-concordant** function on $\mathbf{R}$ is a convex differentiable function that fulfills:

$$|f^{'''}(x)| \leq 2f^{''}(x)^{3/2}.$$

A function defined in $\mathbf{R}^n$ is self-corcondant if in every 1-D ray is self-concordant.

Most functions that appear in applications (linear, quadratic, $f(x) = -\sum_i \log(x_i)$, $f(x) = -\log det(X)$, etc) are self-concordant. If g is self-concordant, then so is $f(x) = g(Ax + b)$.

The very important result by Nesterov and Nemirovskii is that for self-concordant functions the number of iterations for obtaining a solution with error ϵ is

$$C(f(x^{(0)}) - p^*) + \log\log(1/\epsilon),$$

for some constant C (which in many cases is small, e.g. $C = 2$). On the other hand, in most practical cases $\log\log(1/\epsilon) = 4$.

15.7 Complexity of Gradient Descent versus Newton's method

As we have seen, in many situations the number of iterations for Gradient Descent is way larger than for the Newton's method. On the other hand, each step in the Newton method involves the solution of a linear system of equations of dimension n.

Although in theory the number of operations required for solving this equation of of the order n^3, in many practical situations the Hessian matrix has a special structure (e.g it is sparse, banded, arrow etc), and the solution of this system requires a much smaller number of operations.

If the Hessian is too expensive, or singular, then we can use an approximation to $H_f(x)$, leading to **quasi-Newton** methods.

In most practical situations, Newton's method will be the most efficient method for solving the problem.

15.8 Newton method with equality constrained minimization

Assume now the COP

$$min_x f_0(x).$$

$$s.t. \quad Ax = b.$$

The Newton method in this case is equivalent to solve a QP with linear constraint. As we have seen, applying the KKT conditions to this problem will lead to solve KKT equations:

$$\begin{bmatrix} H_{f_0} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \mu^* \end{bmatrix} = \begin{bmatrix} -\nabla f_0 \\ b \end{bmatrix}, \tag{15.8.1}$$

which leads again to a linear system of equations.

15.9 Inequality constrained COPs

Finally we arrive to the general COP

$$min_x f_0(x).$$

$$s.t. \quad f_i(x) \leq 0.$$

$$Ax = b.$$

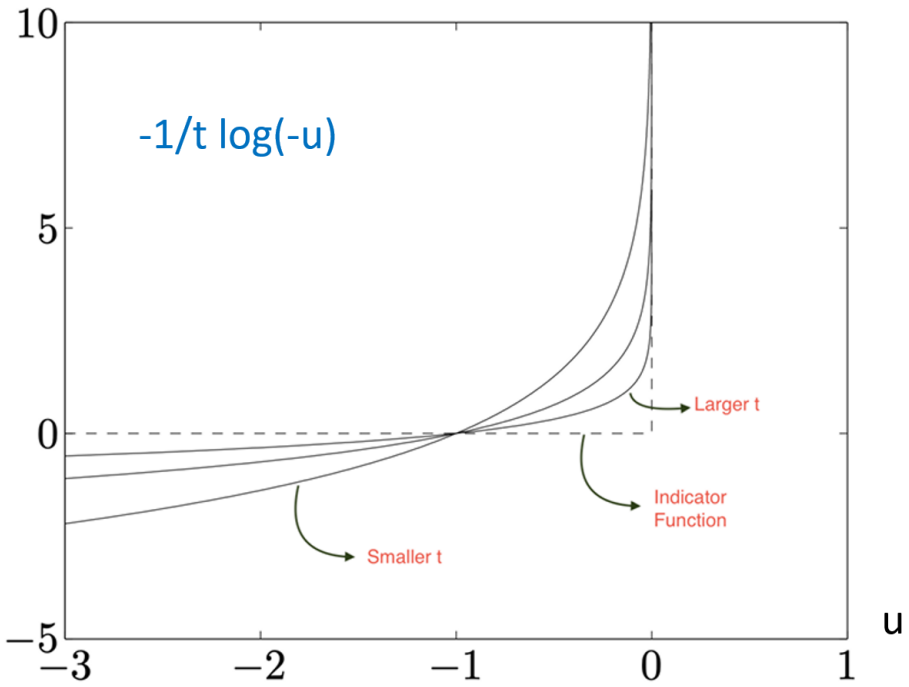
15.10 Logarithmic barrier function and central path

We know that we can transform this problem into a equality constrain COP by using as objective function $f_0(x) + \sum_i I_{f_i(x)\leq 0}$.

We can use the approximation:

$$I_{f_i(x)\leq 0} \approx -\frac{1}{t}\log(-f_i(x)),$$

which is a convex function.



For a given t, we can solve the COP:

$$min_x f_0(x) - \frac{1}{t}\sum_i \log(-f_i(x)),$$

$$s.t. \quad Ax = b.$$

We expect that the convergence to the solution will be long for large values of t. The smaller the value of t, we expect to have a faster solution to our problem,

but a larger error between our approximate solution and the real solution of the problem.

Lets call the optimal point of the previous problem $x^*(t)$. These points are said to belong to the **central path**. For self-concordant functions, it can be proof that we can bound the error as:

$$f_0(x(t)^*) - f_0(x^*) \leq \frac{p}{t},$$

meaning that we can chose the value required final of t.

15.10.1 Following the central path

Leveraging this result, we can set the following strategy for finding a solution: we will start by solving the solution for a small value of t. Then we increase the value of t, e.g. choosing $t^{(n+1)} = \mu t^{(n)}$, and solve the problem using Newton's method, taking the previous solution as initial value. If the two solutions are not far away, we expect that the Newton's methods will work in the quadratic regime, meaning that with a few iteration we will find the new solution. We proceed then by finding these values of the central path until we reach the final value of t.

This strategy works usually very well, meaning that we have a practical method for approaching general COPs.

The figure shows the number of iteration for solve a LP. We can observe that the procedure is pretty insensitive to the actual value of μ.

